

Hands-on Fragility: A Multi-Modal Pseudo-Haptic System for Delicate Interaction in VR

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ABSTRACT

This research demo presents a multi-modal pseudo-haptic system designed to simulate object fragility in Virtual Reality (VR) using bare-hand interaction. While consumer VR lacks granular haptic feedback, our system employs a procedural cracking shader, stress-dependent audio, and a diegetic UI to elicit a sense of "care" in users. A user study ($N = 31$) confirms that these software-driven cues allow users to distinguish between fragility levels and significantly alter their motor behavior. This demonstration allows participants to experience this psychological tension firsthand, providing a novel framework for handling delicate virtual artifacts without specialized hardware.

Index Terms: Pseudo-haptics, Hand Tracking, Material Fragility, Diegetic UI, Multi-modal Feedback.

1 MOTIVATION AND INNOVATION

In the real world, interacting with delicate objects requires a continuous feedback loop of tactile and visual cues. In current consumer VR, the absence of physical resistance often leads to "superhuman" interactions where users inadvertently "crush" virtual objects. While haptic gloves attempt to solve this, their high cost and mechanical complexity limit widespread application [4].

The **novelty** of our system lies in its *software-only* approach to induce **sensory recalibration** [1]. Instead of physical force, we utilize **pseudo-haptics** [3] to create psychological tension. By leveraging *visual dominance*, we demonstrate that calibrated visual and auditory feedback can modulate a user's motor control, inducing cautious behavior equivalent to handling real-world fragile materials.

2 SYSTEM ARCHITECTURE

The system is built for the Meta Quest 3 using Unity. It utilizes controller-less hand tracking to map grip intensity $G \in [0, 1]$.

2.1 Robotic Representation and Diegetic UI

We represent the user's hands as **robotic avatars**. This choice mitigates the "uncanny valley" effect and provides a logical narrative for the integrated sensors. A holographic **diegetic gauge** is attached to the virtual wrist (fig. 1), providing a precise readout of the applied force G relative to the object's breaking point T_{break} .

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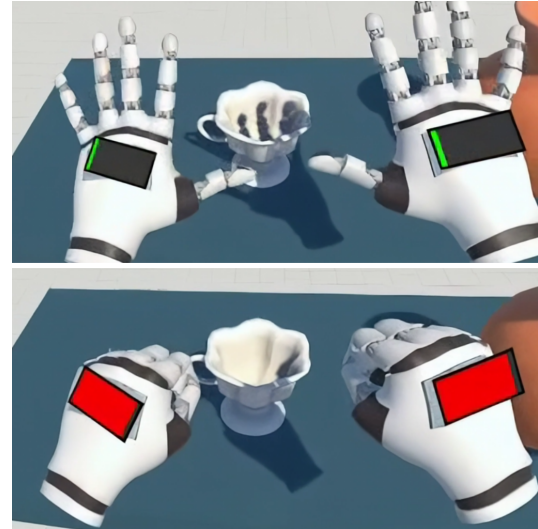


Figure 1: Diegetic UI: The robotic hand integrates a safety gauge providing real-time feedback on the grip force G .

2.2 Procedural Visual Feedback

As G increases, a shader dynamically calculates a Voronoi-based crack pattern. The crack width W is a function of the stress ratio:

$$W = \max\left(0, \frac{G - T_{warn}}{T_{break} - T_{warn}}\right) \quad (1)$$

where T_{warn} is the threshold for visible damage (fig. 2).

2.3 Auditory Synthesis Calculation

The audio engine acts as a proxy for tactile micro-fractures. We modulate the auditory feedback based on the grip's first derivative. The sound volume S_{vol} and pitch S_{pitch} are mapped as follows:

$$S_{vol} = \alpha \cdot \log\left(1 + \left|\frac{dG}{dt}\right|\right), \quad S_{pitch} = \beta \cdot G \quad (2)$$

This logarithmic mapping mimics human perception of loudness, providing a visceral warning when the user squeezes too quickly [2].

3 USER EVALUATION AND RESULTS

A study with 31 participants ($M = 24.2, SD = 4.1$) showed significant differences in perceived fragility across objects ($\chi^2(2) = 50.47, p < .001$). Participants exhibited a **40% reduction in hand velocity** when handling the Test Tube (High Fragility). This behavioral shift proves that multi-modal cues effectively substitute for missing haptic resistance by eliciting a "caution reflex."

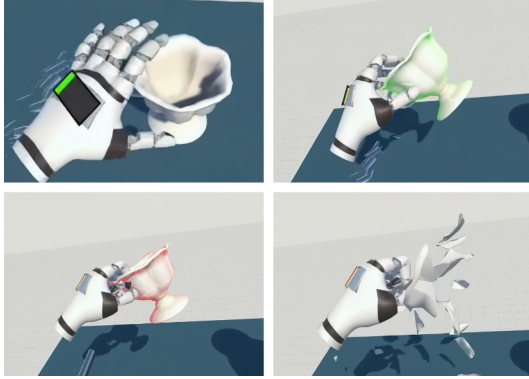


Figure 2: Visual Stress Progression: Real-time procedural cracking on a ceramic object during a high-intensity grasp.

4 DEMONSTRATION SCENARIO

The live demo invites attendees to be "Virtual Conservators."

- **Task:** Transfer three objects of varying fragility into a target zone.
- **Technology Display:** Attendees can toggle an "Ablation Mode" to compare the experience with and without the robotic UI or the sound synthesis, highlighting the importance of multi-modal integration.

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